APPENDIX TO THE SYSTEM OVERVIEW

Compliance Data Store Logic

Every table, Power Query step and formula column of Compliance Data.xlsm as documented by the client, what each one does, where it breaks, and what it means for the build

Source	sources/Wellbeing Dashboard current data template.xlsx, filled in by the client. Tabs 01 Data Sources, 02 Data Size, 03 Checks You Do, 04 Changes You Make (Compliance rows 187–274, all nine columns read)
Covers	Z:\Performance Dashboards and Data\Data Store\Compliance Data.xlsm and the 11 user-input tools that feed it
Related	Compliance_Dashboard_System_Overview.pdf (same folder) and the specification and pre-build PDFs in folders 01–07. Items added to the pre-build checklists cite this document as “DSL §n”
Report date	16 October 2026
VBA status	The template contains **no compliance VBA**. Its only macro (tab 04 row 49) belongs to the Self-Pay Customers tool; see §7

Why this matters. The recordings show people using the tools, but the data store itself was only glimpsed. This template is the client’s own written record of it, so it settles several rules the specs had to infer: what “Issue = 10” means, how Training is judged, what Thresholds 1–5 do and why the store needs “refresh twice”. The client also marks 17 of the 88 rows as “Unclear” or “Unknown”; those become keep-or-retire decisions.

Contents

1. What the Template Says About Compliance 3

2. Data Flow Inside Compliance Data.xlsm 4

2.1 The twelve compliance areas 4

3. Compliance Data Table (the record store) 5

3.1 Snapshot and Last Updated 5

4. Requirements Tables (Power Query and formula columns) 6

4.1 Requirements_Detailed query (rows 202–209) 6

4.2 Requirements query (rows 249–257) 6

4.3 Formula columns added to both tables (rows 210–226 and 258–274) 6

5. Scoring Model (RAG roll-up) 7

5.1 Sub-group RAG (rows 243–244) 7

5.2 Worked example 7

6. The Self-Referencing Query and “Refresh Twice” 8

6.1 Roll-up columns on requirements_Final_Self_Ref (rows 234–248) 8

7. VBA 9

8. Findings 10

9. What This Means for Each Area 11

9.1 Build implications 11

10. Questions for the Client 12

1. What the Template Says About Compliance

Tab	Compliance content	What it tells us
01 Data Sources	Compliance Data (PW & MP): governance platforms (e.g. Datix) plus data requested from contractors, site managers and departments (e.g. Estates); exports, reports and email; mostly monthly, some weekly; owner DPT, tasks rotate	Several people run the same steps in turn, so the rules must live in the system, not in one person’s head
01 Data Sources	User Input Tools: 11 inherited tools in Z:\Performance Dashboards and Data\User input, each updated monthly. Two tables per tool: Names → Compliance Timeframes → data store. Power Query updates the tools; “information is copied into the Compliance Data data store via VBA. No transformation as such is performed”. Knowledge of them is “anecdotal”	Confirms the shared pattern in the Overview §3. The VBA claim conflicts with the recordings (mostly manual paste), see DS-12
02 Data Size	3,874 records; 5–20 new per week; kept 7 years; data back to 2019 with only a few early entries	A small data set: one relational database is enough. 7-year retention becomes a requirement
03 Checks You Do	No automated checks. Errors are spotted by “perusing existing data via these tools”	Every validation in the specs is new; nothing existing to preserve
04 Changes You Make	88 rows: Snapshot, Compliance Data, Last Updated, Requirements_Detailed, compliance_Timeframes_Table, requirements_Final_Self_Ref, Requirements (§2–§6)	The full calculation chain from raw records to the dashboard’s three reference tables

2. Data Flow Inside Compliance Data.xlsm

The client: the file “is its own data store. Data is written to it both manually and via semi-automated input tools… then transformed within the file and then transformed further by multiple Power Queries… to produce three reference tables which then drive the PW and MP Dashboards (format: Power BI Report)” (tab 04 row 187).

Step	Object	Built by	Feeds
1	**Compliance Data** table (compliance_Data_Store_Table): one row per completion record	Rows pasted or written by each tool	Requirements tables (Due Date, Status, Notes)
2	**compliance_Timeframes_Table**: one row per compliance type	Power Query: Table.Combine of 12 tool timeframe tables	Requirements tables; Core Requirement
3	**Requirements_Detailed**: one row per person/site × type	Power Query (11 requirement tables + timeframes join) + formula columns	Power BI
4	**Requirements**: same as 3 with Group 1/2 instead of 3/4	Power Query (12 requirement tables) + formula columns	Power BI; step 5
5	**requirements_Final_Self_Ref**: sub-group and group roll-ups	Power Query reading this same file from Z:\, + formula columns	Power BI
6	**Last Updated**: one row per Group L1–L2	Typed by hand or written by tool VBA	Last Reviewed Date
–	**Snapshot**: weekly records by service, from September 2023	Unknown	Unknown (client: purpose unclear)

2.1 The twelve compliance areas

The Power Query combine lists name the areas. Folder numbers show which ones have been recorded.

Area (table prefix)	Timeframes	Requirements	Requirements_Detailed	Recorded in
estates	✓	✓	✓	—
staff	✓	✓	✓	01
organisation	✓	✓	✓	—
audit_Programme	✓	✓	✓	—
regular_Audit	✓	✓	✓	—
policy	✓	✓	✓	—
medical_Devices	✓	✓	✓	—
incidents	✓	✓	✓	02, 07
Sickness_and_Annual_Leave	✓	✓	✓	—
Survey	✓	✓	✓	05
training	✓	✓	✓	04
complications	✓	✓	**✗ missing**	03

RTT (folder 06) does not appear by name. Its rows must sit inside one of these areas (likely organisation or audit); see DS-11.

3. Compliance Data Table (the record store)

Rows 191–200. Every column is a formula on top of the pasted record columns (Name, Compliance Type, Completion Date, Completion Status, Due Date, Issue, Compliance Group Level 1/2, Compliance Notes).

Column	Rule (plain words)	Client's reason	Our note
Most Recent Record	TRUE when this row has the latest Completion Date for its Unique Lookup and Compliance Type; also TRUE when no row of that pair has a date	Keep the full history while the tools read only the latest record	With no dates every row is "most recent" (DS-01)
Compliant	Most recent rows only. By Compliance Group – Level 2: • Incidents, Feedback, HR Metrics, Complications: Issue = 10 → Compliant • Training: Completion Status not blank → Compliant • all others: Due Date after today, **or blank** → Compliant	Required for reporting	Explains Issue = 10. A blank due date counts as compliant (DS-02); Training ignores expiry (DS-03)
Expires within 3 months	Most recent rows with a due date: before EDATE(TODAY(),3) → "Expires within 3 months"; for **Ongoing Audit** before WORKDAY(TODAY(),10) → "Expires within 2-weeks"	Required for reporting	No lower bound, so already-expired items are labelled "expires" too (DS-04)
Weeks Non Compliant	ROUND((TODAY() − Due Date)/7, 0) when Non Compliant	Required for reporting	Blank due date gives ≈ 6,600 weeks; Issue-based failures with a future due date give negative weeks (DS-05)
Core Requirement	Looks up the type in compliance_Timeframes_Table → "Core" or blank	Required for reporting	Core flag is per type, not per person
Unique Lookup	Name-Compliance Type	Assumed used later	Name text as key (Overview §5)
Unique Lookup 2	Name-Type-Completion Status-Completion Date (most recent rows only)	Assumed used later	No consumer documented
Unique Lookup 3	Unique Lookup (most recent rows only)	Assumed used later	Seen in spec 01 lookups
Unique Lookup 4	Name-Type (most recent rows only)	Assumed used later	**The join key** used by every Requirements lookup
Notes	Looks up Unique Lookup 4 in requirements_Table[Notes]	Required for reporting	Notes live on the requirement, not the record

3.1 Snapshot and Last Updated

Table	Columns	Client's note
Snapshot	Most recent location record (MAXIFS on Date Week Commencing by Service-Completion Status); Most recent service record (by Service Group-Completion Status); the two Unique Lookups	Data dates back to September 2023 (V1 of the tool). "It is unclear what function this data performs now"
Last Updated	Unique Lookup = Compliance Group Level 1-Level 2; one Updated date per group	Written "manually or semi-automatically by each user input tool. Lookup may relate to the VBA contained in these tools"

4. Requirements Tables (Power Query and formula columns)

4.1 Requirements_Detailed query (rows 202–209)

```
Source            = Table.Combine({estates_, staff_, organisation_, audit_Programme_, regular_Audit_, policy_,
                      medical_Devices_, incidents_, Sickness_and_Annual_Leave_, Survey_, training_ ...requirements_Table})
Merged Queries    = Table.NestedJoin(Source, {"Compliance Type"}, compliance_Timeframes_Table, ..., JoinKind.LeftOuter)
Expanded          = Compliance Group 3, Compliance Group 4, Threshold 1..5
Filtered Rows     = keep rows where Compliance Group 3 <> null
Renamed Columns   = Group 3 -> Compliance Sub Groups, Group 4 -> Compliance Groups, Notes -> Requirements Notes
Sorted Rows       = Compliance Groups desc, Required (T/F) desc
Reordered Columns = Name, Compliance Type, Required (T/F), Core Requirement, Requirements Notes, Managed By,
                    Unique Lookup, Compliance Timeframe, Onboarding, Groups, Sub Groups, Threshold 1..5
```

4.2 Requirements query (rows 249–257)

The same chain, with three differences: it also combines **complications_requirements_Table**; it expands **Compliance Group 1 and 2** (not 3 and 4); and both filter steps are each `true`, so they keep every row. There are therefore two group hierarchies (1/2 and 3/4) feeding different dashboard tables (DS-08).

4.3 Formula columns added to both tables (rows 210–226 and 258–274)

Column	Rule	Client's reason
Compliance Group and Sub Group; …, Sub Group and Type	Text keys "Group-Sub Group" and "Group-Sub Group-Type"	Next columns and Power BI visuals
Due Date	From the store by Unique Lookup 4. No record → text "No record"; date 0 → "No expiry date set"	Monitor and chase; feeds Power BI
Compliance Status	From the store's Compliant. **No record → "Non Compliant"**	Monitor and chase; feeds Power BI
Notes	From the store's Compliance Notes	Monitor and chase
Compliance Completion Status	From the store's Completion Status; no record → "Missing"	Unclear whether the dashboard uses it
Compliance / Compliance (Core)	1 if Compliant, 0 if not; blank unless Required = TRUE **and Active = TRUE** (and Core = "Core" for the Core column)	Used in the percentage
% Compliant (and Core) per type	AVERAGEIFS of Compliance for the Group-Sub Group-Type key, on its first row only	Feeds Power BI
Type RAG Score (and Core)	% ≥ Threshold 1 → 1; ≥ T2 → 0.8; ≥ T3 → 0.6; ≥ T4 → 0.4; ≥ T5 → 0.2; else 0	"Used later in a compliance rank – unclear on reporting purpose"
Type Encoded	Character codes of the type name joined by spaces	Unclear
Type Rank (Base / Base 2 / Base 3 / Final)	Rank within sub group by Core RAG score, ascending (1 = worst); ties broken by adding encoded-name digits ÷ 10^{16}; Final = MAX per type	Unclear

5. Scoring Model (RAG roll-up)

Scores run from 0 to 1. The dashboard gauges are coloured from these scores against Thresholds 1–5 of each compliance type (set per type in the tools' Compliance Timeframes tables).

Level	Percentage	RAG score
Type	Average of 1/0 over required, active rows	Threshold bands 1 / 0.8 / 0.6 / 0.4 / 0.2 / 0
Sub group	Average of type percentages (Onboarding excluded)	Special rules for Incidents and HR Metrics; otherwise see 5.1 (Accreditation excluded)
Group	Average of sub-group percentages	Threshold bands against the group's average thresholds (Accreditation Prep excluded; Core version excludes "Accreditation")

5.1 Sub-group RAG (rows 243–244)

Sub group	Rule, checked in this order
Incidents	1. Any type at RAG 0 whose name contains neither "SPC" nor "Moderate" → **0** 2. Exactly one "Moderate Harm" at RAG 0 → **0.5** 3. Two or more "SPC" types at RAG 0 → **0.5** 4. One "SPC" type at RAG 0 → **0.8** 5. Otherwise → **1**
HR Metrics	Two or more SPC types at RAG 0 → 0.5; one → 0.8; otherwise 1
All others	Gate: average type RAG minus a penalty for types at 0 % (0.3 for one, 0.6 for two, 0.9 for three, 1 for four or more). If the result is below 0 → **0**. Otherwise the sub group's % against the average Thresholds 1–5 → 1 / 0.8 / 0.6 / 0.4 / 0.2 / 0

Note. The penalty only acts as a gate: it can force 0, but it never lowers a score that passes. Two failing Moderate Harm types fall through to the SPC rules and can score 1 (DS-07).

5.2 Worked example

A sub group with four types at 100 %, 100 %, 50 % and 0 %, all with thresholds 1 / 0.9 / 0.8 / 0.7 / 0.6:

- Type RAG scores: 1, 1, 0, 0 (50 % is below Threshold 5). Average 0.5.
- One type at 0 % → penalty 0.3. 0.5 − 0.3 = 0.2, not below 0, so the gate passes.
- Sub-group % = 62.5 %, which is ≥ Threshold 5 (0.6) but < Threshold 4 (0.7) → **RAG 0.2**.
- With a second type at 0 % the penalty would be 0.6 and the gate would force 0.

6. The Self-Referencing Query and “Refresh Twice”

```
Source         = Excel.Workbook(File.Contents("Z:\Performance Dashboards and Data\Data Store\Compliance Data.xlsm"), null, true)
Navigation     = Source{[Item="requirements_Table",Kind="Table"]}[Data]
Changed Type   = 26 typed columns
Removed Other  = keep 29 columns (incl. Active, Due Date, Compliance Type Rank (Final))
Appended Query = Table.Combine({#"Removed Other Columns", requirements_Table_Self_Reference})
Changed Type 1 = Compliance Type Rank (Final) as Int64
```

- The query reads **the saved copy of its own file** from Z:\, so it sees the values from the last save, not what is on screen. The formulas must recalculate and the file be saved before the roll-ups are right: this is why the recordings refresh twice.
- If the file is open elsewhere or the path changes, the query fails, which matches the “UNABLE TO REFRESH – linked workbook” message.
- It then appends `requirements_Table_Self_Reference`, whose contents are not documented. If it holds the previous output, rows may be doubled (DS-09).
- In the new system all of this becomes one calculation step after load; no file reads itself.

6.1 Roll-up columns on requirements_Final_Self_Ref (rows 234–248)

Column	Rule	Client’s reason
Sub Groups % Compliant (and Core)	Average of type % per Group-Sub Group, first row only; Onboarding excluded	Feeds Power BI
Groups % Compliant (and Core)	Average of sub-group % per group, first row only	Feeds Power BI
Sub Groups Encoded, Rank Base–Final	Same tie-break ranking as types, within each group; Onboarding and Accreditation Prep excluded	Unclear
Sub Groups RAG (and Core)	§5.1	Unclear
Groups RAG (and Core)	Threshold bands on group %	Unclear
Last Reviewed Date	Last Updated[Updated] for Group-Sub Group	Data quality
Column1	Average of Threshold 3 per Group-Sub Group	“Currently unknown what threshold references refer to”

7. VBA

No compliance VBA is in the template. The word appears in the Compliance rows only in the Last Updated note (tab 04 row 201). The one macro supplied (tab 04 row 49) belongs to the Self-Pay Customers tool. It is the only example of how the client's tools write to a store, so it shows the likely pattern of the compliance tools.

```
Private Sub Workbook_Open()   ' opens the store workbook when the tool opens
  Workbooks.Open "Z:\Invoices & Scorecards\Main Data Store\Self Pay Reporting\Self Pay Customers Store.xlsx"
Sub copyData()
  New_Referrals.Unprotect : New_Referrals.Calculate
  PD_Customers.AutoFilter Field:=14, Criteria1:="Copy - new customer"
  copy visible A2:V  ->  PasteSpecial xlPasteValues below the last row of 'PowerDiary Customers'
  clear filter : Protect : wb2.Save : Debug.Print "Done"
```

Pattern	Risk in the compliance tools
Append-only, chosen by a helper flag column	No check against the store, so a wrong flag or a second run duplicates rows
Fixed positions (A:V, field 14)	Adding or moving a column silently copies the wrong data
Hard-coded Z:\ paths	Breaks when a drive mapping or folder changes
No log, no rollback, no user stamp	No audit trail of who wrote what and when

Still required: a .bas export of every module in the 11 tools and in Compliance Data.xlsm, and a list of which tools use a macro and which are pasted by hand (DS-12).

8. Findings

ID	Finding	Evidence	Effect
DS-01	Most Recent Record is TRUE for every row of a person/type when none has a Completion Date	Row 191	Several “latest” rows; XLOOKUP returns the first, which may not be right
DS-02	Default rule counts a blank Due Date as Compliant	Row 192	Missing data shows green
DS-03	Training is Compliant whenever Completion Status is filled, regardless of expiry	Row 192	Lapsed training may show green; confirm against spec 04
DS-04	“Expires within 3 months / 2 weeks” has no lower bound	Row 193	Overdue items are also labelled as expiring
DS-05	Weeks Non Compliant uses Due Date even where status comes from Issue	Rows 192, 194	≈ 6,600 or negative weeks
DS-06	Missing record → “Non Compliant”; Due Date column mixes dates with “No record” text	Rows 212–213	Correct for chasing, but the column type is mixed
DS-07	Incidents sub-group rule: two failing Moderate Harm types are not caught by rule 2	Row 243	Score may be 1 instead of 0.5 or 0
DS-08	Two group hierarchies: Requirements uses Group 1/2, Requirements_Detailed Group 3/4	Rows 204, 251	Same data, two groupings in the dashboard; ties to spec 01 decision D-02
DS-09	Self-referencing query plus an undocumented self-reference append	Rows 228–233	“Refresh twice”, refresh errors, possible doubled rows
DS-10	Complications missing from Requirements_Detailed but present in Requirements and Timeframes	Rows 202, 227, 249	Complications may be absent from one dashboard table
DS-11	RTT (folder 06) is not a named area	Rows 202, 227	Its group must be confirmed to know which rule applies
DS-12	Template says tools copy via VBA; recordings show mostly manual paste; no compliance VBA supplied	01 B12, row 49	Write logic still unknown
DS-13	Exclusion names differ: “Accreditation” vs “Accreditation Prep”	Rows 239, 243–246	One of them probably never matches
DS-14	Formula anchors differ between columns (Q$1:Q1, Q$1:Q2, J$1:J3)	Rows 243–246	If the range includes the current row, the first-row test always fails and the column is blank; check in the live file
DS-15	First-row tests use fixed column letters (J, Q, S)	Rows 218, 234, 236	Inserting a column breaks the roll-ups
DS-16	Rank tie-break adds 14 digits ÷ 10^{16} to an integer rank	Rows 224, 240	Beyond double precision, so ties can survive
DS-17	17 rows marked Unclear/Unknown: Snapshot, Encoded, Rank columns, roll-up RAGs, Column1	Column “Why You Do It”	Keep-or-retire decision needed

9. What This Means for Each Area

Folder	Level 2 rule that applies	Extra roll-up rule	Ask added to the pre-build PDF
01 Staff Management	Default: Due Date after today or blank	—	Blank due date = compliant? (DS-02)
02 / 07 Incidents	Issue = 10	Incidents sub-group rule (§5.1)	Confirm the rule and the Moderate Harm gap (DS-07)
03 Complications	Issue = 10	Default gate	Add to Requirements_Detailed? (DS-10)
04 Mandatory Training	Completion Status not blank	Default gate	Should expiry count? (DS-03)
05 Survey Feedback	Issue = 10	Default gate	Confirm Issue = 10 for SPC results
06 RTT	Depends on its group (not named)	Depends on its group	Which area and Level 2 group? (DS-11)

Every folder also gets three shared asks: the tool’s VBA export, the missing-record and Active rules, and the keep-or-retire list.

9.1 Build implications

- **Rules engine:** one table of compliance rules keyed by Level 2 group (Issue score, completion status or due date), versioned, replacing the SWITCH in row 192.
- **Scoring service:** type → sub group → group roll-up with thresholds per type, the Incidents and HR Metrics rules as named, tested rules, and the exclusions as configuration.
- **Explicit states:** “No record”, “No expiry date set” and “Missing” become status values, not text in a date column.
- **Latest-record view:** a database view replaces Most Recent Record and Unique Lookups 1–4; ties resolved by a record ID, not by position.
- **No self-reference:** calculations run once after each load, and the publish date replaces the Last Updated table.
- **Parity test:** recompute today’s scores from a copy of Compliance Data.xlsm and compare type, sub-group and group results before go-live.
- **Sizing:** about 4,000 records growing by up to 20 a week, kept for 7 years.

10. Questions for the Client

These also appear, by area, in each folder’s pre-build checklist.

✓	ID	Question for the client	Owner
☐	DS-02	Should a blank due date count as compliant?	Business owner
☐	DS-03	Should Training lapse at expiry, or stay compliant once completed?	Business owner
☐	DS-04	Should overdue items be excluded from “expires within”?	Business owner
☐	DS-07	What should two failing Moderate Harm types score?	Incident governance lead
☐	DS-08	Which group hierarchy (1/2 or 3/4) does each dashboard page use? Are both needed?	BI owner
☐	DS-09	What is requirements_Table_Self_Reference?	Tool owner
☐	DS-10	Should Complications be in Requirements_Detailed?	Tool owner
☐	DS-11	Which area and Level 2 group does RTT belong to?	Tool owner
☐	DS-12	Export all VBA modules; which tools use a macro to write to the store?	IT / tool owner
☐	DS-13	Is the exclusion “Accreditation” or “Accreditation Prep”?	Business owner
☐	DS-14	Are the Core sub-group and group RAG columns populated in the live file?	Tool owner
☐	DS-17	Keep or retire Snapshot, the Encoded and Rank columns, and Column1?	Business owner + BI owner